

for T&M instruments."

Recently, the government has announced the 3G spectrum auctions. The T&M markets can now expect a lot of activity in the network optimisation sectors. "The requirements will be for receivers, scanners, test mobiles, replay and analysis and optimisation software. There may also be an increase in demand for hand-held test instruments to be used in field trials," says Mohan.

"The T&M industry must also gear up for other areas like the biomedical sector, where measurements of optical and magnetic properties are as important as those of currents, voltages and frequencies. The rapidly emerging field of nanotechnology is another potential area," he adds. These numerous areas of applications of T&M instruments have created a virtually evergreen market in India.

Trends in T&M

According to Viraj Pradhan, communication sales manager, Asia Pacific, Tektronix, one of the recent trends in T&M pertaining to the telecom domain, is a heightened demand for protocol analysis-based network optimisation solutions. This trend springs from the new radio access network environment, including cell breathing and power management in universal mobile telecommunication systems (UMTS) and with the introduction of multi-input, multi-output (MIMO) and beamforming that need continuous monitoring approach in order to capture the rapid variances in the network.

"However, the T&M equipment manufactured in India need to adapt to the new standard requirements (data speed, encryption, dynamic resource allocation, etc) of modern systems," says Joseph.

There is a surge in instances of testing moving from the bench to onsite locations—be it cell towers, power substation yards or network control racks. "Full-featured, compact and rugged T&M tools are therefore



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the current rage," states Joseph.

The latest advancements in T&M are mostly pertinent to protocol testing, Wi-Fi test systems, subscriber identity module (SIM) card tests and network testers. "We have solutions for wireless radio communication sets which are used in the defence sector and feature encryption and frequency hopping to facilitate secure communications. We are now focusing on testers for Wi-Fi testing," claims Sabapathi.

Today more than 90 per cent of all commercial wideband code division multiple access (W-CDMA) networks globally support high-speed downlink packet access (HSDPA) and many are being upgraded to high-speed uplink packet access (HSUPA). Mobile data traffic is escalating, and network operators will soon have to upgrade their 3G/high-speed packet access (HSPA) networks to either HSPA+LTE or both.

"To meet the operators' production plans for HSPA+ and LTE, all major chipset, mobile and infrastructure manufacturers currently have projects in operation. Rohde & Schwarz has extended its entire existing HSPA portfolio to support HSPA+ and to offer a market-leading portfolio for LTE frequency-division duplex (FDD) and time-division duplex (TDD)," says Mohan.

Tektronix Communications has also created a new monitoring solution in order to meet the challenges faced by the manufacturers through the LTE access network architecture. "A new probe for protocol analysis at the air interface has been developed, allowing seamless monitoring and testing of all wireless and wireline LTE network interfaces," discloses Pradhan.

Why manufacturing is not happening?

Despite the tremendous opportunities for T&M instruments in India in various domains, it is unfortunate that domestic manufacturing is still not picking up. "A majority of the